INITIALS: NT PETA

STUDENT NUMBER: 223570943

BTO-1 Lab Pack (Foundations)

Contents

BTO-1 Lab Pack (Foundations) – Documentation

Title: Blue Team Lab Setup, Sysmon Integration, and Microsoft Sentinel Analytics Validation

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Objective

To build foundational Blue Team skills by deploying and validating Microsoft Sentinel analytics, signals, and visualizations using a simulated enterprise data environment. This includes configuring Sysmon, Data Collection Rules (DCRs), and securely transferring Sysmon logs to the Sentinel workspace.

1. Signal Onboarding & Health

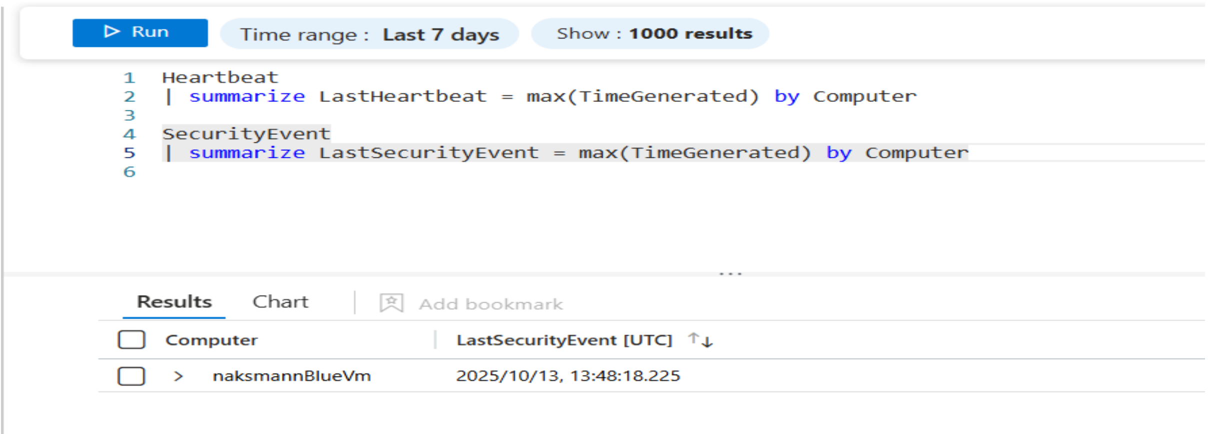
Goal

Enable Microsoft Sentinel and verify that telemetry is arriving from connected sources.

Steps

1. Open Microsoft Sentinel and connect it to your Log Analytics Workspace.
2. In Data Connectors, enable:
   - Azure Monitor Agent (AMA)
   - Security Events (Windows Event Logs)
3. Run the queries below to confirm heartbeat and security events are fresh:

```
Heartbeat
| summarize LastHeartbeat = max(TimeGenerated) by Computer
SecurityEvent
| summarize LastSecurityEvent = max(TimeGenerated) by Computer
```



4. Created a Health Workbook showing the freshness of these two sources.

Verification

- Each connected machine appears with a recent heartbeat timestamp.
- SecurityEvent logs are received within the past few minutes.

2. Windows Security via AMA

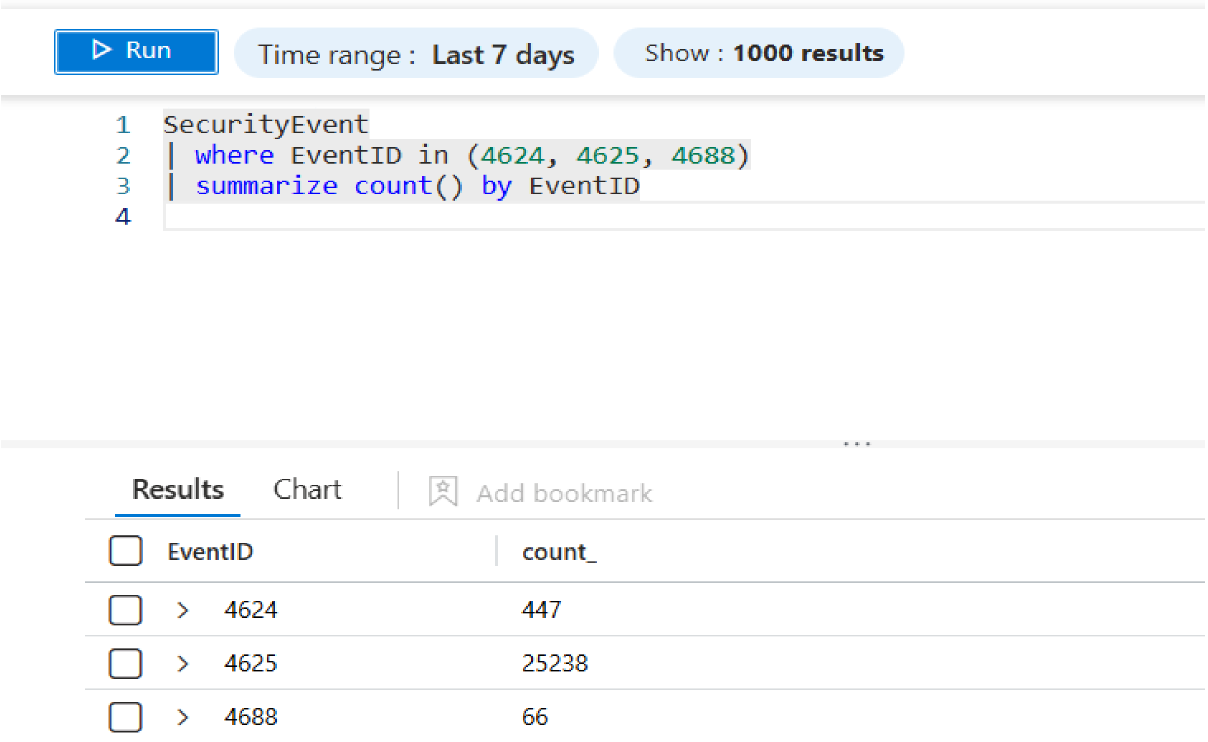
Goal

Validate collection of native Windows Security logs (Event IDs 4624, 4625, 4688).

Steps

1. Create a Data Collection Rule (DCR) for your Windows hosts.
    - Log Category: SecurityEvents
    - Destination: Your Sentinel workspace
2. Verify EventIDs 4624 (Logon), 4625 (Failed logon), and 4688 (Process Creation):

```
SecurityEvent
| where EventID in (4624, 4625, 4688)
| summarize count() by EventID
```



Verification

- Count for each event ID > 0 confirms data is flowing successfully.

3. Sysmon + DCR (Sysmon Log Transfer Integration)

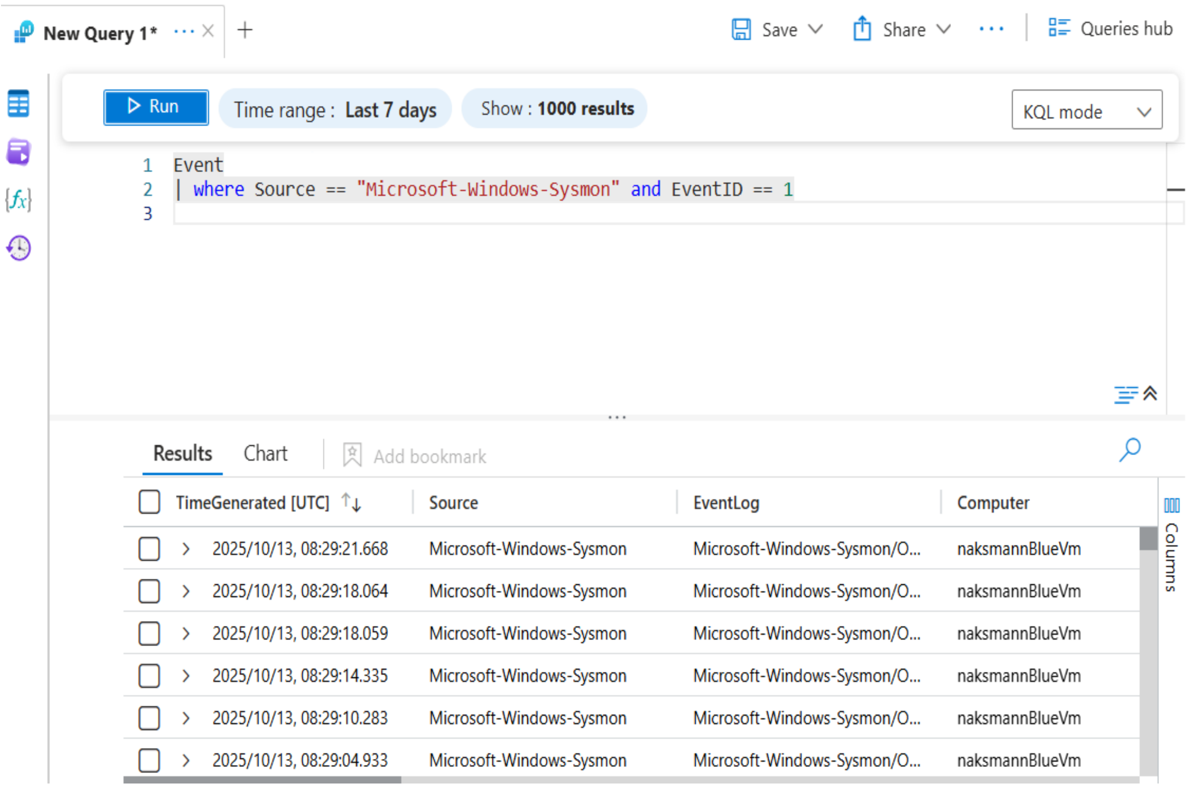
Goal

Enhance telemetry with Sysmon for detailed process and network monitoring, and successfully send Sysmon logs to the DCR you created for Microsoft Sentinel.

Steps

1. Installed Sysmon on my lab endpoint using the following command:
2. sysmon.exe -i sysmonconfig.xml
3. Created a new DCR to collect Sysmon logs:
   - Source: Microsoft-Windows-Sysmon/Operational
   - Destination: Your Sentinel workspace
4. Linked my endpoint to this DCR to ensure all Sysmon logs are being forwarded.
5. Verified Sysmon log transfer by running the query below:

Event
| where Source == "Microsoft-Windows-Sysmon" and EventID == 1



Verification

- Results show process image paths such as C:\Windows\System32\notepad.exe
- Confirms Sysmon telemetry is being collected and successfully transferred via your DCR.

4. KQL Essentials

Goal

Learn to query, filter, and summarize data using KQL.

Starter Queries

a. **Using where:**

SecurityEvent
| where EventID == 4625

b. **Using summarize:**

SecurityEvent
| summarize FailedLogons = count() by Computer

c. **Using bin:**

SecurityEvent
| where EventID == 4625
| summarize count() by bin(TimeGenerated, 1h)

5. OOB Rule Enablement

Goal

Enable Out-of-the-Box (OOB) analytics rules to automatically detect threats.

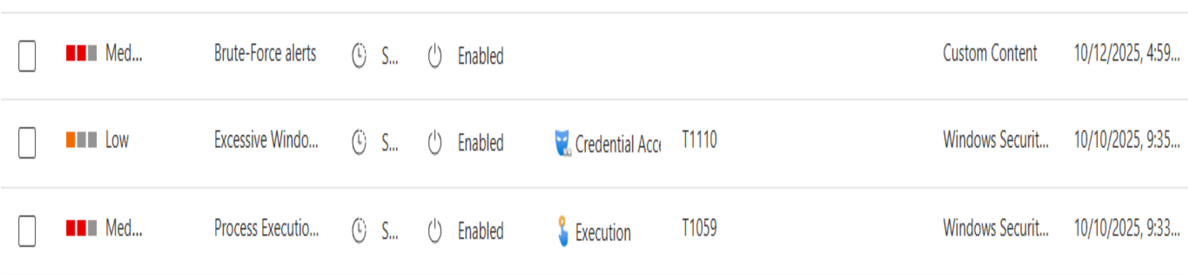
Steps

1. Went to:

   Sentinel
   Analytics
   Rule Templates.

2. Enabled the following example rules:
   - Brute Force Attack against Windows Local Accounts
   - Suspicious Process Execution (Sysmon)

3. Confirm their status = Enabled.



Verification

- Alerts appear in Incidents after simulating failed logons or suspicious process activity.

6. Workbook 101

Goal

Built visual dashboards using KQL queries.

Steps

Went to:

Workbooks(+ Add new ,Blank Workbook).

Added these visualizations:

a. Failed vs Successful Logons

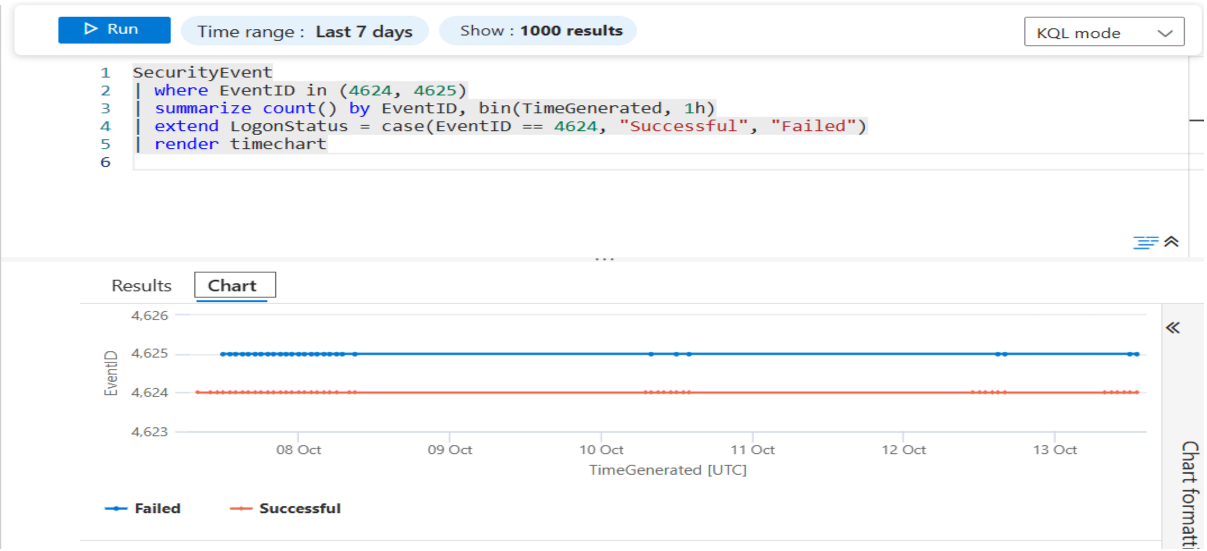
SecurityEvent

| where EventID in (4624, 4625)

| summarize count() by EventID, bin(TimeGenerated, 1h)

| extend LogonStatus = case(EventID == 4624, "Successful", "Failed")

| render timechart



b. Sysmon – Top N Processes

Event

| where Source == "Microsoft-Windows-Sysmon" and EventID == 1

```
| extend parsed = parse_xml(EventData)
| extend EventData = parsed.EventData.DataItem.EventData
| mv-expand Data = EventData.Data
| extend Name = tostring(Data['@Name']), Value = tostring(Data['#text'])
| where Name == "Image"
| summarize count() by Image = Value
| top 10 by count_
| render barchart
```

Verification

- Workbook shows a time chart for logons and a bar chart for top processes.

7. Watchlist 10

Goal

Import high-value user or asset lists and use them for monitoring and correlation.

Option A — High-Value Users

CSV Used:

```
SearchKey,AssetRole
naksmann_03,IT Administrator
naksmann_04,C-Level Executive
naksmann_05,Finance Manager
```

Steps

1. Upload CSV in Sentinel (Configuration, Watchlists, Add new watchlist.)
   - Name: HighValueUsers
2. Use the query below to correlate events:

```
let hv = _GetWatchlist('HighValueUsers');
SecurityEvent
| where TimeGenerated > ago(1d)
| where EventID in (4624, 4625, 4688)
| extend AccountShort = tostring(split(Account, '\\')[-1])
| lookup kind=leftouter hv on $left.AccountShort == $right.SearchKey
| summarize cnt = count() by AccountShort, AssetRole, bin(TimeGenerated, 1h)
| sort by cnt desc
```

Result: Displays logon and process events for high-value users.

Option B — High-Value Assets

CSV Used:

```
SearchKey,AssetRole
DC01,Domain Controller
FIN-SRV01,Finance Server
HR-SRV02,HR Server
```

Query

```
let hv = _GetWatchlist('HighValueAssets');
SecurityEvent
| where EventID in (4625, 4688)
| lookup kind=leftouter hv on $left.Computer == $right.SearchKey
| summarize cnt = count() by coalesce(Computer, 'unknown'), bin(TimeGenerated,
1h)
```

Result: Displays failed logons or process activity for critical servers.

Conclusion

This lab successfully demonstrated the deployment and integration of Microsoft Sentinel with various telemetry sources.
By setting up Sysmon and connecting it to the custom DCR, detailed endpoint activity (process creation, network events) was successfully captured and sent to Sentinel.